PHASE2 Data Quality Report

This report evaluates the reliability, completeness and consistency of hospital operational and billing data before deploying predictive moels. The analysis focuses on missing values, outliers, and business-rule validation.

Missing Value Analysis

Key fields evaluated were:

- approved_amount
- payment_days
- length_of_stay_hours

Findings:

Column	Missing Count	% Missing
approved_amount	1318	5.27%
payment_days	790	3.16%
length_of_stay_hours	0	0.00%

Business Interpretation:

- approved_amount (5.27% missing) – Mostly related to pending or rejected claims. Which suggests a slight risk for revenue analysis but manageable.
- payment_days (3.16% missing) – Missing values correspong to unpaid or recently submitted claims. Minimal impact on cash flow.
- Length_of_stay_hours(0% missing) – Complete data ensures operational metrics are reliable.

Outlier Detection

Outliers were detected using thr IQR method on numeric columns:

Column	Number of Outliers
billed_amount	373
payment_days	509
length_of_stay_hours	256

- billed_amount: Extreme values likely represent ICU cases. Important for revenur risk analysis. It may need special treatment in models.
- payment_days: High values indicate delayed payments, thus affecting cash flows.
- length_of_stay_hours: Outliers correspond to prolonged cases and operational resource planning should consider these.

Business Rule Validation

Length of stay hours has 0 invalid values

Payment days has minor anomalies and should be reviewed

Approved amount exceeding billed amount is none

The dataset is sound and operationally reliable. Primary risks are minor missing financial data and outliers in billing, payment and length of stay metrics. With standard preprocessing and outlier handling, the data is suitable for predictive model and financial analysis.